

KUET Kinship

Architecture, Developer Flow, Public User Flow & Limitations

Build: kuet_kinship_beta_build_101.apk

1. Overview

KUET Kinship is an Android-based campus utility application intended to bring useful KUET academic and campus resources into one place. The current beta build focuses on academic and campus features such as course selection, examinations, lecture notes, and rooms/laboratories.

2. High-Level Architecture

The application is organized as a set of feature-specific screens. The main activity provides the central application structure, while separate Activities and Fragments handle individual features.

KUET Kinship Android App	
MainActivity	
HomeFragment / Main Dashboard	
Academic Year Selection → Term Selection → Course Selection	Examination Exam Information
Learning Lecture Notes	Campus Rooms & Labs → Room Navigation

Main Components

Component	Role
MainActivity	Main application entry point and navigation structure
HomeFragment	Central dashboard
YearSelectionFragment	Select academic year
TermSelectionFragment	Select term/semester
CourseSelectionFragment	Select/access course-related information
ExamAndStuffActivity / ExamsActivity	Exam-related information
LectureNotesActivity	Lecture-note resources
RoomsAndLabsActivity	Room and laboratory information
RoomNavigateActivity	Navigation for a selected room

3. Developer-Oriented Application Flow

This section describes the internal flow of the application from a developer's perspective. It is intentionally separate from the simple public-facing usage instructions below.

1. Application Entry: The Android application starts through MainActivity, which establishes the main screen/navigation structure.

2. Dashboard Layer: HomeFragment serves as the central dashboard from which the implemented feature modules can be reached.
3. Academic Navigation: The academic flow is divided into YearSelectionFragment, TermSelectionFragment and CourseSelectionFragment, allowing the application to narrow the user's selection step by step.
4. Feature Modules: Exam, lecture-note, and room/lab functionality is separated into dedicated Activities. This keeps feature-specific UI and logic modular.
5. Room Navigation: RoomsAndLabsActivity handles room/lab information and RoomNavigateActivity handles the navigation experience for a selected room.
6. UI Binding and Asynchronous Components: The APK includes generated binding classes for Activities and Fragments and Kotlin Coroutine-related components, supporting structured UI access and asynchronous work.

4. How the App Works — General User Flow

This section is written for general users, not developers. The goal is to explain the app in a simple 'tap → choose → get information' style.

Finding Course Information

Open KUET Kinship → go to the academic/course section → select your year → select your term/semester → select your course → view the available course information.

Checking Exam Information

Open KUET Kinship → go to the exam section → choose the relevant academic information → open the available exam information → check the details.

Reading Lecture Notes

Open KUET Kinship → tap the lecture-notes section → choose the relevant academic/course information → open the available note → read the material.

Finding a Room or Lab

Open KUET Kinship → go to Rooms & Labs → find/select the required room or lab → view its information → use the navigation option to help reach the selected location.

In One Simple Example

Suppose you want to find information about a course: Open the app → choose your year → choose your term → choose the course → see the available information. That's the basic idea: the user selects what they need step by step, and the app shows the relevant information.

5. Implemented Scope of the Current Beta Build

- Home/dashboard structure.
- Academic year, term and course selection flow.
- Exam-related module.
- Lecture-notes module.
- Rooms and laboratories module.
- Room navigation module.

6. Limitations

Beta-stage limitation: This is a beta build, so some functionality may still be under development or subject to change.

Data freshness: The usefulness of academic, exam, course, note and room information depends on the correctness and freshness of the available data.

Limited scope: The current build is focused on academic and campus utilities rather than being a complete university management system.

Navigation dependency: Room navigation depends on accurate and sufficiently complete room/location information.

Network/data-source dependency: Static APK inspection cannot confirm the exact backend behavior of every feature. Features that retrieve external information may depend on connectivity and the underlying data source.

No guarantee of real-time information: Schedules, room allocations and other campus information can change, so information may become outdated if it is not synchronized regularly.

Device compatibility: Performance and UI behavior can vary across Android versions, screen sizes and hardware.

Future scalability: A larger user base or much larger data volume may require stronger backend, database, caching and synchronization infrastructure.

7. Strengths

- Feature-based modular structure makes the app easier to organize and maintain.
- Academic information is broken into simple selection steps for users.
- Dedicated modules separate courses, exams, notes and rooms/labs.
- The architecture provides a reasonable foundation for adding more campus features in future versions.

8. Conclusion

KUET Kinship provides a centralized way for KUET students to access academic and campus-related resources. The current beta build focuses on course selection, exam information, lecture notes, and room/lab utilities. Its modular architecture makes future expansion possible, while the main areas for improvement are data freshness, real-time synchronization, navigation accuracy, broader functionality and scalability.

Important note: The architecture and component descriptions in this document are based on static inspection of the supplied APK. The exact server-side architecture, backend APIs, database schema and runtime network behavior cannot be confirmed from the APK alone.